Practical-09

Aim: To implement a simple planning problem using STRIPS or Goal-Stack Planning in a Blocks World, logistics, or robot navigation environment to achieve a given goal through a sequence of actions.

CODE:

```
initial_state = {"At_A"}
goal ="At_C"
actions = [
    {
        "name":"Move A to B",
        "pre":{"At_A"},
        "add":{"At_B"},
        "delete":{"At_A"}
    },
    {
        "name":"Move B to C",
        "pre":{"At_B"},
        "add":{"At_C"},
        "delete":{"At_B"}
    }
]
        def find_action(goal):
          for action in actions:
            if goal in action["add"]:
              return action

          return None
```


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```
def make_plan(goal,state,plan):

   if goal in state:
    return True
   action = find_action(goal)

   if action is None:
    return False

   for condition in action["pre"]:
    if not make_plan(condition,state,plan):
      return False
   state.difference_update(action["delete"])
   state.update(action["add"])
   plan.append(action["name"])

   return True
```

```
state = initial_state.copy()
plan =[]

make_plan(goal,state,plan)

print("Plan:")
for step in plan:
  print(step)
```

OUTPUT:

```
Plan:
Move A to B
Move B to C
```

Conclusion:

Thus, a simple planning problem was successfully implemented using STRIPS or Goal-Stack Planning. The system identified the required actions and arranged them in the correct sequence to achieve the goal. This practical helped us understand AI planning techniques and automated problem-solving.